Knot-music: Spatial Mixing by Untangling Wired Earphones

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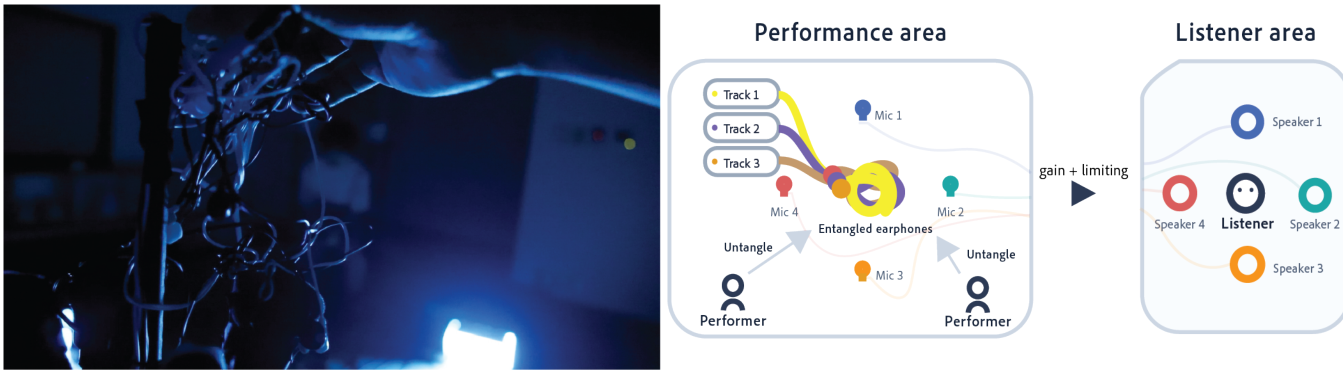

Fig. 1. Left: Knot-music in performance, with the three sounding earphones entangled between the performers' hands. Right: the signal path. Three tracks play continuously, one through each earphone. Four microphones stand around the working area, and each feeds, with gain and limiting, the loudspeaker holding the corresponding position in a four-channel array set close to a single listener. The microphones pick up the sound leaking from the earpieces together with the friction, impact and contact of handling, so the untangling is reproduced along with the tracks. Because the correspondence is positional, moving an earphone within the working area moves it across the four channels at the listener, who sees the hands throughout.

We present Knot-music, a live performance in which entangled earphones play different tracks while two performers untangle them. Knot-music treats the wired earphone as turntablism treats the record player. Four microphones stand around the working area, and each is routed to a loudspeaker holding the corresponding position in a four-channel array set close to the listener's ears. Therefore, where an earphone sits determines how strongly it is reproduced by each of the four loudspeakers. In many instruments the surface a player touches and the body that produces the sound are separate parts, linked by a mapping. In Knot-music they are the same object. The entangled earphones constrain the hands, carry their own playback signals, and stay where the hands leave them, so a change to the constraint is also a change to the sound, and is heard when it is made.

Additional Key Words and Phrases: spatial audio, sound performance, tangible interaction, physical constraints, turntablism

1 Introduction

Turntablism treats recorded tracks as material for live performance, repurposing a playback device as an instrument [Smith 2000]. Knot-music takes the same approach to the wired earphone. An earphone is a loudspeaker small enough to be carried in one hand, so moving it moves a source of sound. Earphones also tangle, in pockets and bags, and untangling them is a task most people have carried out many times without instruction. Knot-music puts three of them together: each plays a different track, and the three stay entangled while the performers work. The entanglement couples the sources mechanically, since moving one displaces another and loosening one crossing changes the tension elsewhere, so the current configuration conditions each action and each action produces a new configuration.

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Knot-music uses this changing configuration as the structure of a spatial-audio performance. We contribute an implemented performance in which the arrangement that constrains the hands is the arrangement of the sounding bodies themselves. Because the constraint and the sources are one arrangement, the performers act directly on the spatial result, and a change to the constraint is heard when it is made.

2 Related Work

Knot-music focuses on the materiality of earphones, specifically their entanglement. Cables and knots have served as musical interfaces before, in each case by being read. Cord UIs formulated the cord as an input surface, where tying or pinching a cable controls the device at its end, with a sensor reading the cable and a metaphor of flow making the control legible [Schoessler et al. 2015]. Cadavid ties knots in conductive cord whose changing resistance is read as they are made, sending control values to effects on pre-recorded tracks [Cadavid 2020]. Kasich, surveying uses of knot theory in computer music, proposes both knotted trajectories for virtual sources and a rope-shaped controller that identifies its configuration by counting crossings [Kasich 2024]. In all of them a knot or a cable yields a value, and the sound it affects is made elsewhere. Kasich notes in that survey that no existing work uses knots physically for spatialization. In Knot-music, the cable carries the sound it is used to shape, so what the hands work on is what sounds, and moving it is the spatial operation.

The same identity appears where loudspeakers themselves are moved. In Monahan's Speaker Swinging, three performers swing loudspeakers overhead on their cables, and the motion itself produces the Doppler shift and phasing that make up the piece [Monahan 1982]. What the performers hold is what sounds, as here. The difference is the cable: there it tethers one hand to one speaker, while here it also attaches the three sources to each other, so moving one earphone moves the others.

That mutual attachment is what the hands meet, and it works as a constraint. Magnusson describes digital instruments as systems of constraints, physical and compositional, that enclose a space of possible musical action, and these constraints are settled when the instrument is designed, apart from the body that makes the sound [Magnusson 2010]. Korg's Kaoss Pad is designed on this principle: one touch surface is coupled to several parameters at once, so a small movement produces a change the player has not specified in detail. In Knot-music the coupling is mechanical rather than mapped, and the constraints are produced in the course of playing: each manipulation leaves a configuration that conditions the next, so the performers work locally while the overall task of untangling stays fixed.

3 Artwork and System

The performance takes place in a room of approximately 5 m × 5 m. Three MP3 players play a different track continuously through each of three wired earphones, which are entangled into a single bundle. Four microphones stand around the working area. Each microphone input is sent to the loudspeaker holding the corresponding position in a four-channel array placed close to a single listener and away from the microphones, with gain and limiting applied. Because each input stays assigned to its own output, a change in captured level changes the four-channel balance directly. The microphones pick up the tracks leaking from the earpieces together with the friction, impact and contact sounds of handling. The listener hears this reproduction along with sound arriving directly from the working area, which stays in view throughout.

Before each performance the cables are tangled again. One performer begins alone, so the coupling among the three earphones is heard on its own first. A second joins partway through, and the bundle then couples the performers as well. A performance lasts about five minutes.

4 Interaction and Sonic Structure

Handling the bundle changes two things at once: the arrangement of crossings and tensions, and the position of the three sounding earphones. The crossings decide how motion travels between them. As they come undone the earphones hold each other less, so a pull that would have moved all three at the start moves only one by the end. Untangling gives the performance an overall aim, and the order of steps is left to the performers. They act on whatever is in front of them, and what they do remains in the bundle.

The cables that constrain the hands are the cables that carry the signals, so the microphones pick up the manipulations along with the tracks. Untangling has a sound of its own, made of friction, impact and contact, and it arrives alongside the music. Position reaches the listener as well: each microphone feeds its own loudspeaker, so an earphone lifted and carried across the working area moves across the four channels as it goes, and the balance among the three tracks shifts with it. The listener sees the hands while hearing all of this, so the movement that made a sound is visible as it is heard.

5 Discussion and Limitations

Turntablism reorganizes recorded material by operating on the medium that plays it, and the operation is heard because the hand and the sound share a surface. A scratch is the sound of a record being handled, and it is treated as music; the practice began in misuse, and the industry later built equipment for it [Smith 2000]. Knot-music keeps that arrangement, altering the spatial relation among three sources, and it inherits the same treatment of contact: the friction and impact of untangling belong to the piece.

Knot-music also narrows what can be controlled separately. The performers loosen and tighten, and what follows includes the balance of three tracks, the position of each source, and the sound of the contacts their hands make, including the ones nobody intended. None of them can be reached on its own, since one movement shifts the balance, the positions and the contact sounds together, so the performers find out what a movement does by making it. Playing the piece is a matter of searching rather than aiming [Magnusson 2010].

The bundle keeps no record of what was intended. *Knot* and *tangle* name a difference it does not hold: a knot is one someone made, a tangle is one that happened. With two performers, what one placed on purpose reaches the other as something that simply happened. Neither can read the other's intention out of the bundle, so the search extends from the material to the partner: each finds out what the other did through the process of untangling.

What listeners make of it we have examined only informally. We have shown the work at an exhibition, performing it ten times, once for each of ten listeners. Listeners reported that the sound of the cables being untangled became part of what they were listening to rather than an interruption of it, and that the closeness of the loudspeakers gave the reproduction a particular intimacy. These reports speak to the audibility we claim: a change to the constraint reached the listener as it was made. How the spatial arrangement is perceived remains to be examined. Establishing it would show that the listener, and not only the performers, encounters the bundle as constraint and source at once.

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